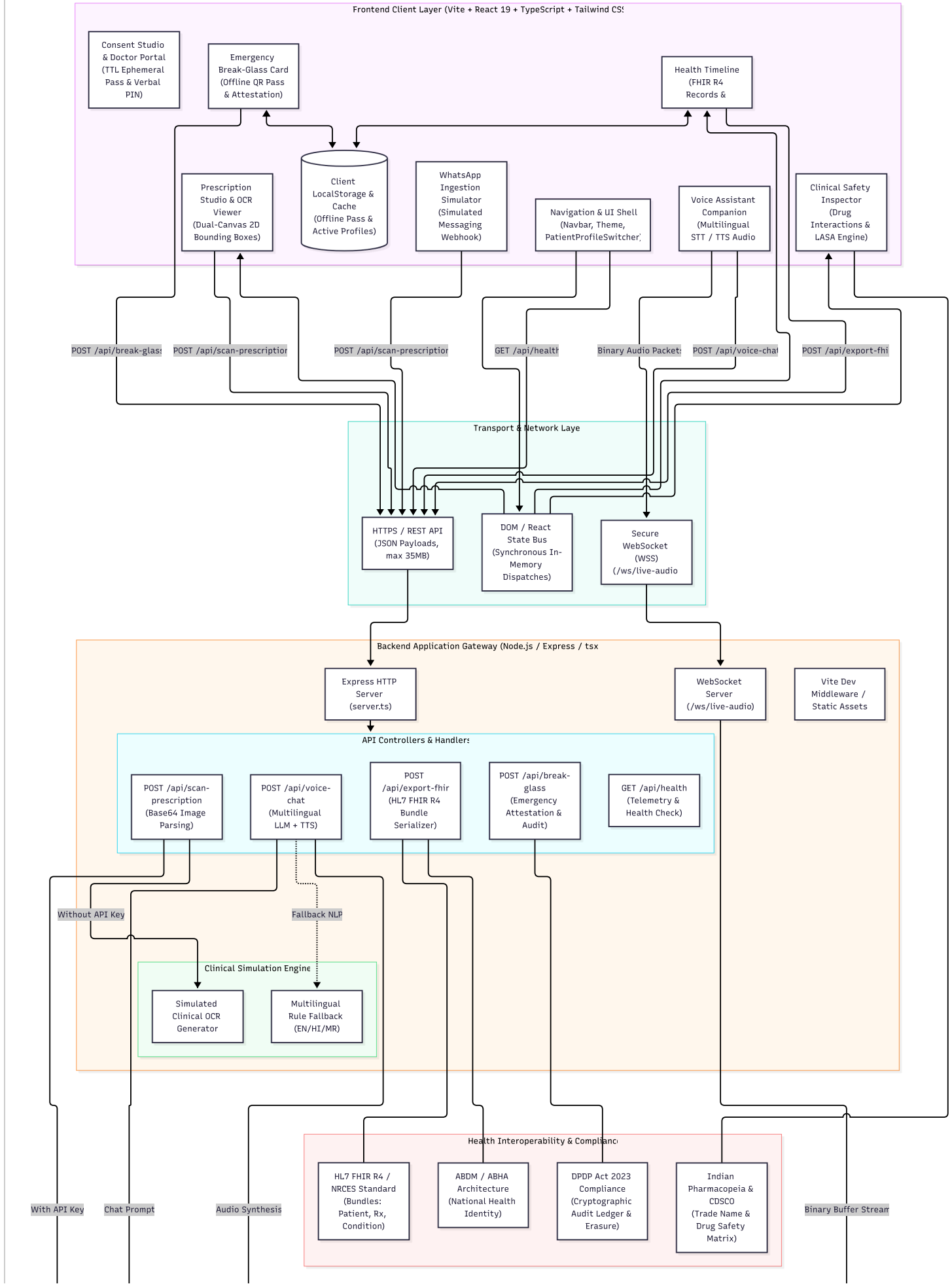
ARCHITECTURE SPECIFICATION · EDITION 2026

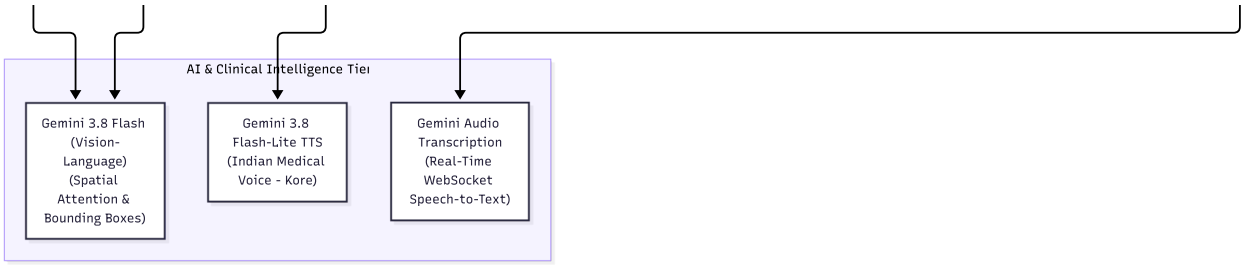
# MediScript: Stack Communication Architecture

Complete technical blueprint of application topography, multimodal AI pipelines, real-time protocols, and regulatory security boundaries.

# 1. High-Level Stack Architecture

End-to-end integration topology between the client workspace, the Node.js Express Gateway, the Google Gemini Multimodal Vision & Speech tier, and national health standards (ABDM / NRCES / DPDP 2023).

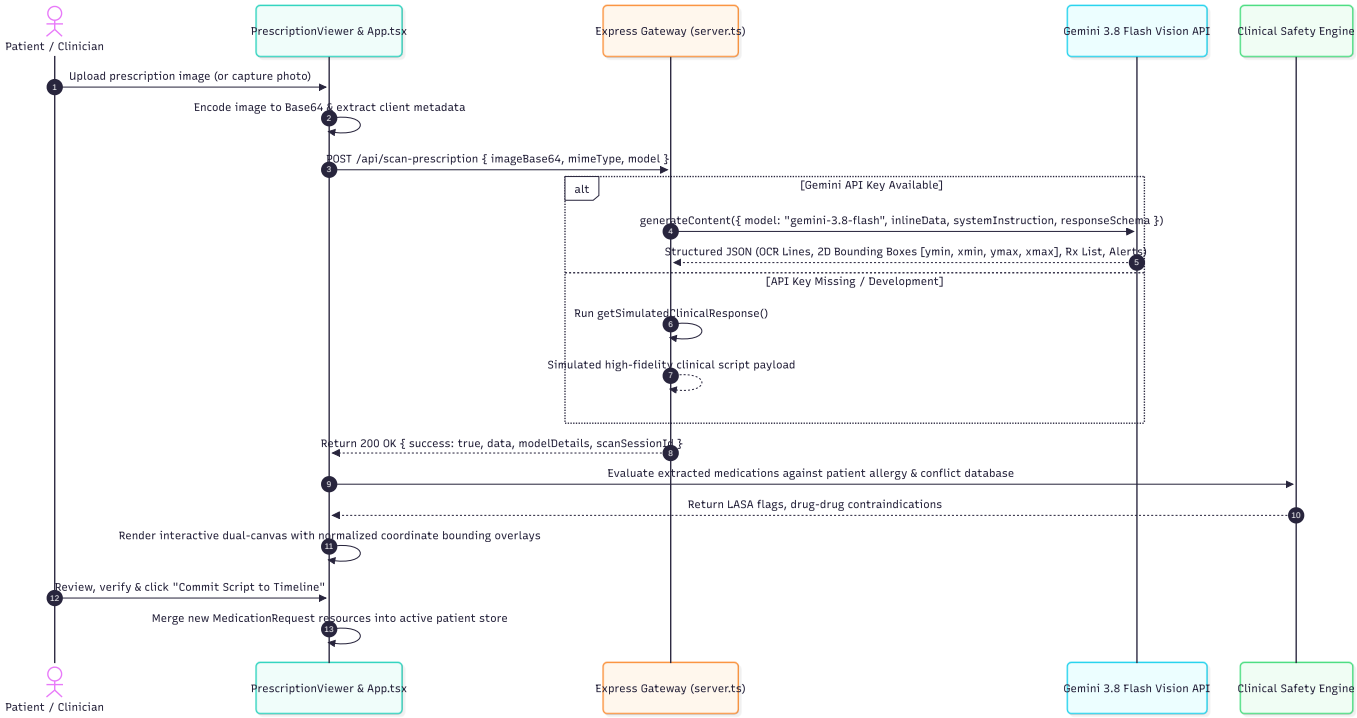

Frontend Client Layer (Vite + React 19 + TypeScript + Tailwind CS
Consent Studio & Doctor Portal (TTL Ephemeral Pass & Verbal PIN)
Emergency Break-Glass Card (Offline QR Pass & Attestation)
Health Timeline (FHIR R4 Records &
Client LocalStorage & Cache (Offline Pass & Active Profiles)
Prescription Studio & OCR Viewer (Dual-Canvas 2D Bounding Boxes)
WhatsApp Ingestion Simulator (Simulated Messaging Webhook)
Navigation & UI Shell (Navbar, Theme, PatientProfileSwitcher)
Voice Assistant Companion (Multilingual STT / TTS Audio
Clinical Safety Inspector (Drug Interactions & LASA Engine)
POST /api/break-glass
POST /api/scan-prescriptior
POST /api/scan-prescriptior
GET /api/health
Binary Audio Packet
POST /api/voice-chat
POST /api/export-fhi
Transport & Network Laye
HTTPS / REST API (JSON Payloads, max 35MB)
DOM / React State Bus (Synchronous In-Memory Dispatches)
Secure WebSocket (WSS) (/ws/live-audio
Backend Application Gateway (Node.js / Express / tsx
Express HTTP Server (server.ts)
WebSocket Server (/ws/live-audio)
Vite Dev Middleware / Static Assets
API Controllers & Handlers
POST /api/scan-prescription (Base64 Image Parsing)
POST /api/voice-chat (Multilingual LLM + TTS)
POST /api/export-fhir (HL7 FHIR R4 Bundle Serializer)
POST /api/break-glass (Emergency Attestation & Audit)
GET /api/health (Telemetry & Health Check)
Without API Key
Fallback NLP
Clinical Simulation Engine
Simulated Clinical OCR Generator
Multilingual Rule Fallback (EN/HI/MR)
Health Interoperability & Compliance
HL7 FHIR R4 / NRCES Standard (Bundles: Patient, Rx, Condition)
ABDM / ABHA Architecture (National Health Identity)
DPDP Act 2023 Compliance (Cryptographic Audit Ledger & Erasure)
Indian Pharmacopeia & CDSCO (Trade Name & Drug Safety Matrix)
With API Key
Chat Prompt
Audio Synthesis
Binary Buffer Stream


AI & Clinical Intelligence Tier
Gemini 3.8 Flash (Vision-Language) (Spatial Attention & Bounding Boxes)
Gemini 3.8 Flash-Lite TTS (Indian Medical Voice - Kore)
Gemini Audio Transcription (Real-Time WebSocket Speech-to-Text)

# 2. Communication Sequences & Protocols

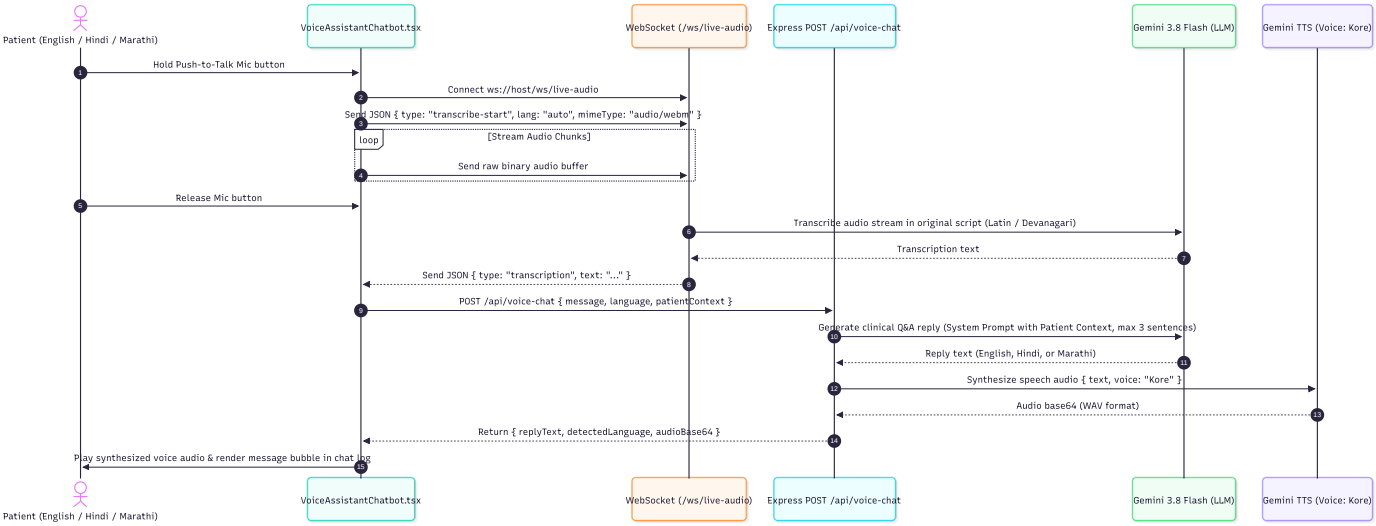
## 2.1 Multimodal Prescription OCR Pipeline

Direct upload, 2D normalized coordinate box extraction `[ymin, xmin, ymax, xmax]`, token confidence scoring, and pharmacological safety review.



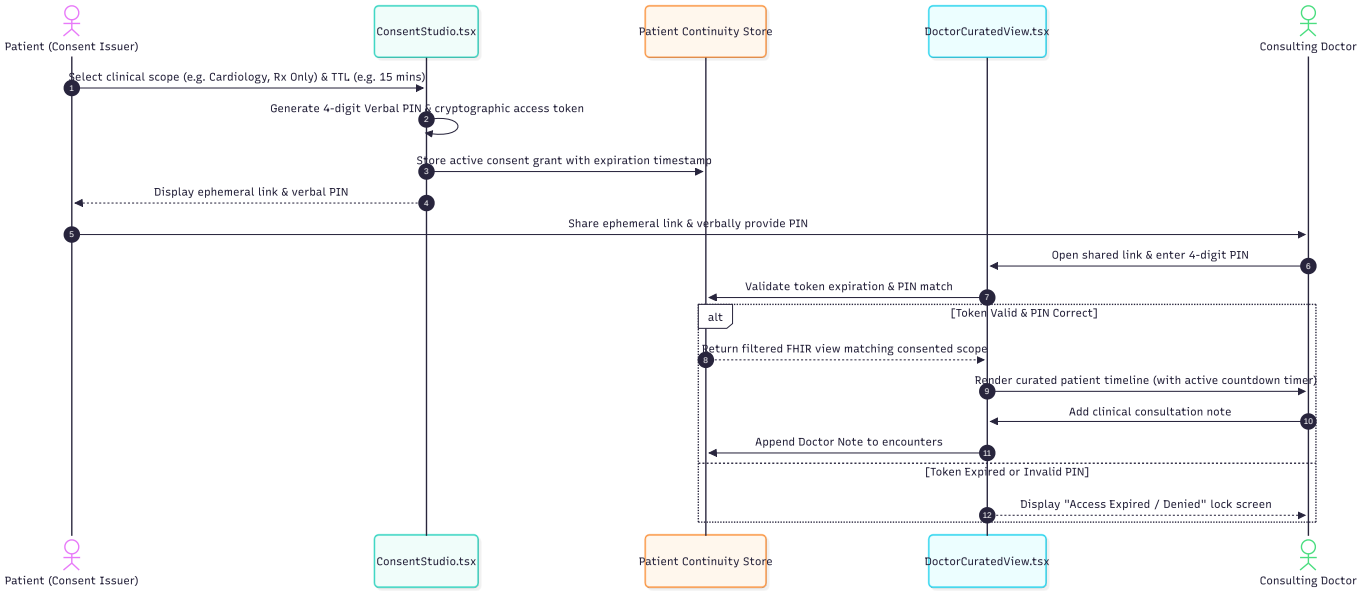

## 2.2 Push-to-Talk Multilingual Voice Assistant

Real-time binary audio stream over WebSockets with Gemini speech-to-text transcription and natural TTS voice playback in English, Hindi, and Marathi.



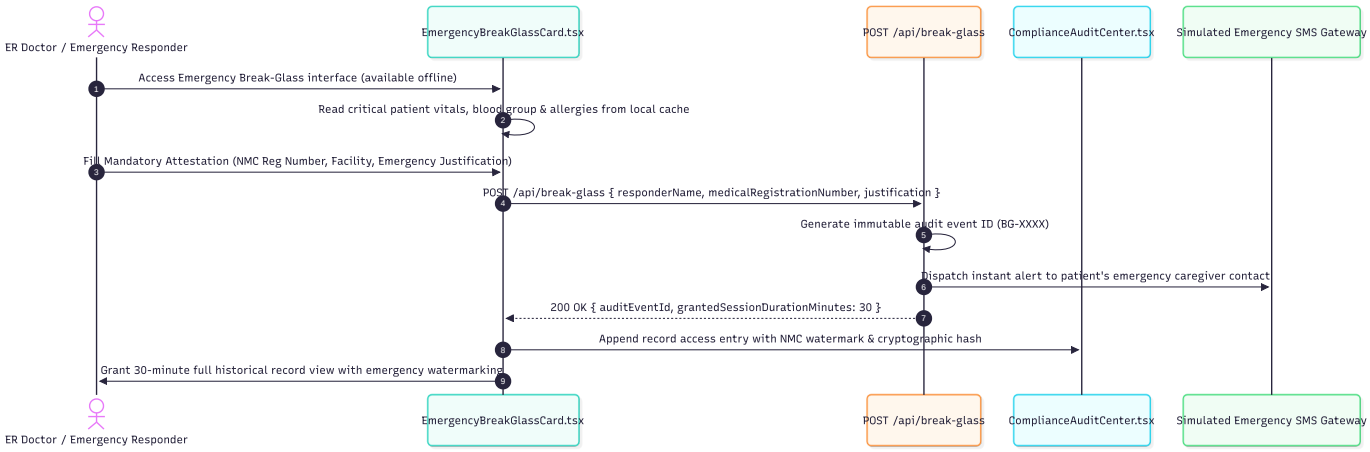
## 2.3 Ephemeral Doctor View & Consent-Gated Access

ABDM & DPDP-compliant time-to-live (TTL) links, verbal 4-digit PIN verification, scoped clinical data filtering, and zero-knowledge doctor curation.




## 2.4 Break-Glass Emergency Offline Pass

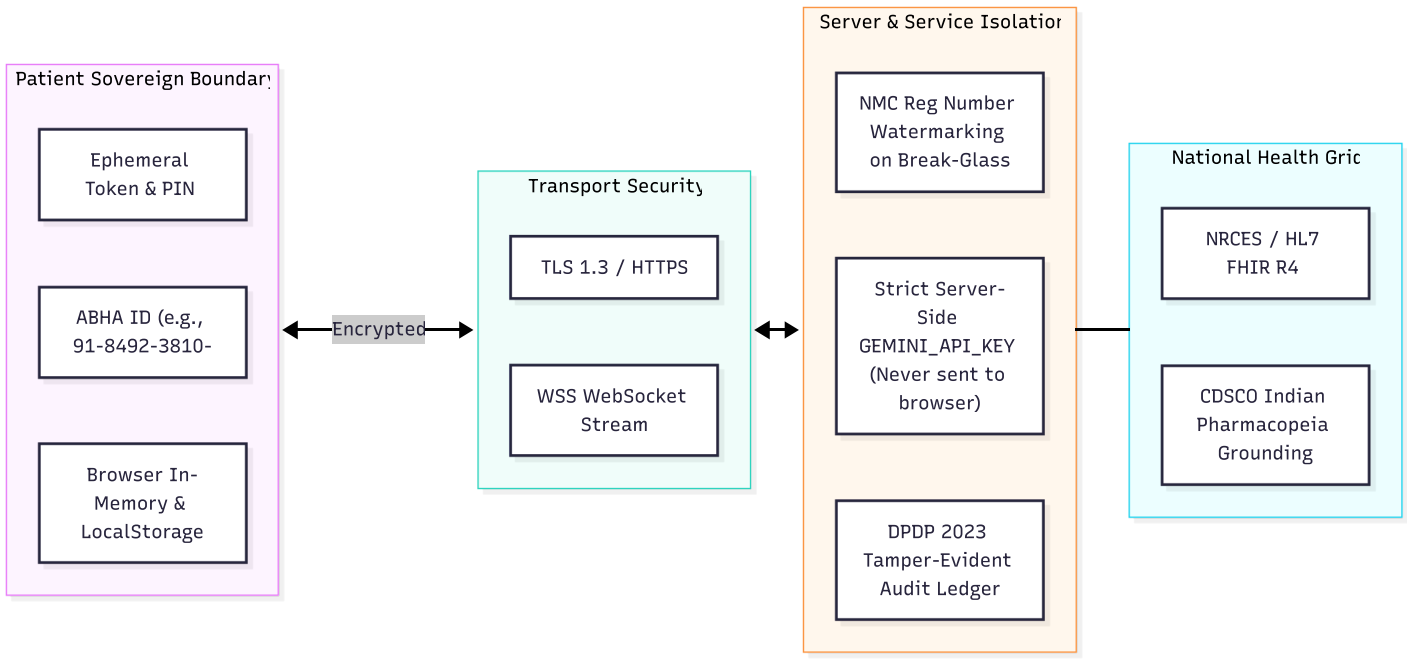
Offline zero-connectivity emergency card with mandatory National Medical Commission (NMC) registration attestation, cryptographic auditing, and instant caregiver SMS alerting.



## 3. Communication Protocols & Payload Matrix

| Channel | Endpoint | Protocol | Payload Format | Typical Size | Latency Budget | Purpose |
|---|---|---|---|---|---|---|
| **OCR Ingestion** | `/api/scan-prescription` | HTTPS POST | JSON (Base64) | 2 MB – 30 MB | 800ms – 2500ms | High-res Rx image transcription & 2D coordinate box parsing |
| **Voice Chat** | `/api/voice-chat` | HTTPS POST | JSON | < 10 KB | 500ms – 1200ms | Conversational clinical assistant query & TTS audio generation |
| **Voice Streaming** | `/ws/live-audio` | WSS Binary | Binary Buffer | ~4 KB / frame | < 150ms | Real-time push-to-talk speech-to-text transcription |
| **FHIR Export** | `/api/export-fhir` | HTTPS POST | JSON | 20 KB – 200 KB | < 100ms | ABDM / NRCES compliant HL7 FHIR R4 Bundle packaging |
| **Break-Glass Pass** | `/api/break-glass` | HTTPS POST | JSON | < 5 KB | < 200ms | Clinician registration attestation & SMS notification trigger |
| **Health Telemetry** | `/api/health` | HTTPS GET | JSON | < 1 KB | < 20ms | Server readiness, active Gemini model & telemetry probe |

## 4. Security, Identity & Regulatory Boundaries




- **Zero-Trust Server Boundary:** `GEMINI_API_KEY` is loaded strictly server-side in `server.ts` and is never delivered to client code.
- **Offline Disaster Survivability:** Critical emergency vitals, blood type, and allergy alerts reside in encrypted browser cache for instant emergency retrieval during zero network conditions.
- **DPDP Act 2023 Sovereignty:** Purpose-limited ephemeral consent, automatic TTL expiration, and cryptographic event auditing.

---

MediScript Healthcare Continuity Platform · Architectural Documentation & Distribution Build